

Carsten Wulff

Analog designer, programmer, educator and group manager

carsten@wulff.no | [Homepage](#) | [CV](#)

Work experience

Nordic Semiconductor	2015-now	Group Manager, Wireless Recruited 20+ people Active in Bluetooth special interest group core specification working group Analog design nRF53 Tool development (Python/C)
	2014-2015	Team Lead Analog design nRF91 Tool development (Perl)
	2011-2014	Senior R & D Engineer Analog design nRF52 Tool development (Perl)
	2008-2011	R & D Engineer Analog design nRF51 Tool development (Perl)
NTNU	2014-2017	Post.Doc A Compiled 9-bit 20-MS/s 3.5-fJ/conv.step SAR ADC in 28-nm FDSOI for Bluetooth Low Energy Receivers , IEEE Journal of Solid-State Circuits
	2004-2008	Ph.D. student Efficient ADCs for nano-scale CMOS technology
	2001-2003	Teaching assistant Planning and execution of transfer of the CAD system from HP-UX to Linux Development av Next Generation Lab, an online electronics lab. Linux administration
Wulff Consulting	2000-2010	Board member and consultant Consulting on .NET and C#
Interconsult group	1999-2000	Summer intern, part time Programming in Visual Basic, Active Server Pages, and PHP.

Lecture experience

2010-now	Guest lectures at NTNU every year, some of the presentations can be found at http://www.wulff.no/presentations
2006	TFE4200, Analog integrated circuits, responsible for frequency response, noise and CMOS modelling

Programming

I've had a love of programming since my first real summer internship back in 1999. Over the years I've programed in Visual BASIC, PHP, Ruby, Perl, C#, SKILL, Ocean, Verilog-A, C++, BASH, AWK, VHDL, SPICE, MATLAB, ASP, Java, C, SystemC, Verilog, and probably a few I've forgotten. These days, it's usually Python, C and C++.	
2019-now	Python frontend to ciccreator – Python
2009-now	Rapid layout generation tool for SAR ADCs – C++
2017	Power meter blink to kWh using nRF52 – C/Python
2015	Ocean script support package – Perl
2004	SystemDotNet.Report – C#
2005	SystemDotNet: mixed-signal, discrete time, event driven simulator – C#
2003	Time Tracker 2 – C#
2001	NextGenLab.Chart – C#

Education	2004-2008	NTNU, Ph.D
	1997-2002	NTNU, Master of Science
Publications		Carsten Wulff on Google Scholar Carsten Wulff on IEEE Xplore
Books	2003	Tor A. Fjeldly, Michael S. Shur "Lab on the Web: Running Real Electronics Experiments via the Internet" Chapter 4: Carsten Wulff, Trond Ytterdal, Thomas Aas Sæthre, Arne Skjevlan, "NEXT GENERATION LABORATORY - A solution for remote characterization of analog integrated circuits" , ISBN: 0-471-41375-5, Wiley, 2003
Journal Papers	2017	Wulff, C; Ytterdal. T. A Compiled 9-bit 20-MS/s 3.5-fJ/conv.step SAR ADC in 28-nm FDSOI for Bluetooth Low Energy Receivers , IEEE Journal of Solid-State Circuits, Vol: 52, Issue: 7, July 2017
	2010	Wulff, C; Ytterdal, T. "Comparator-based switched-capacitor pipelined analog-to-digital converter with comparator preset, and comparator delay compensation" , Analog Integrated Circuits and Signal Processing, December 2010 , © Springer
	2009	L. Lewyn, T. Ytterdal, C. Wulff, and K. Martin, "Analog Circuit Design in Nanoscale CMOS Technologies" , Proceedings of the IEEE , Vol: 97 , No: 10, 2009 DOI: 10.1109/JPROC.2009.2024663 © IEEE
	2009	Wulff, C; Ytterdal, T. "Resonators in Open-Loop Sigma-Delta Modulators" , Circuits and Systems I: Regular Papers, IEEE Transactions on, Vol. 56, No. 10, 2009, DOI: 10.1109/TCSI.2009.2015211 © IEEE
	2008	Wulff, C; Knauserud, Ø; Ytterdal, T. "Switched Capacitor Analog Modulo Integrator For Application In Open Loop Sigma-Delta Modulators" , Analog Integrated Circuits and Signal Processing, February 2008 , © Springer
Conference Papers	2019	Garvik,H; Wulff, C; Ytterdal,T A 68 dB SNDR Compiled Noise-Shaping SAR ADC With On-Chip CDAC Calibration ,IEEE Asian Solid-State Circuits Conference (A-SSCC), 193-194
	2017	Garvik,H; Wulff, C; Ytterdal,T An 11.0 bit ENOB, 9.8 fJ/conv.-step noise-shaping SAR ADC calibrated by least squares estimation IEEE Custom Integrated Circuits Conference (CICC), 1-4
	2016	Wulff, C; Ytterdal, T. "A compiled 3.5fJ/conv.step 9b 20MS/s SAR ADC for wireless applications in 28nm FDSOI" , 42nd European Solid-State Circuits Conference, 2016
	2016	Garvik,H; Wulff, C; Ytterdal,T Noise transfer functions and loop filters especially suited for noise-shaping SAR ADCs , IEEE International Symposium on Circuits and Systems (ISCAS), 1034-1037
	2009	Wulff, C.; Ytterdal, T. "CBSC pipelined ADC with comparator preset, and comparator delay compensation" . Proc. NORCHIP 2009 © IEEE.
	2008	Ytterdal, T; Wulff, C "On the Energy Efficiency of Analog Circuits in Nanoscale CMOS Technologies", International Conference on Microelectronics (ICM2008). Invited paper.© IEEE.
	2008	Wulff, C.; Ytterdal, T. "Design and behavioral simulation of comparator-based switched-capacitor circuits" . Proc NORCHIP 2008.© IEEE. Visual supplements
	2007	Wulff, C.; Ytterdal, T. "Design Of A 7-bit, 200MS/s, 2mW Pipelined ADC With Switched Open-Loop Amplifiers In A 65nm CMOS Technology" . Proc NORCHIP 2007. ISBN 1-4244-1517-9 © IEEE. Visual supplements
	2006	Wulff, C.; Knauserud,Ø; Ytterdal, T. "Analog Modulo Integrator For Use In Open-Loop Sigma-Delta Modulators" . Proceedings of NORCHIP 2006. © IEEE. Visual supplements
	2005	Lillebrekke, C. ; Wulff, C.; Ytterdal, T. "Bootstrapped Switch In Low-Voltage Digital 90nm CMOS Technology" . Proceedings of NORCHIP 2005, © IEEE
	2005	Berntsen, Ø. ; Wulff, C.; Ytterdal, T. "High Speed, High Gain OTA in a Digital 90nm CMOS Technology" . Proceedings of NORCHIP 2005, © IEEE
	2005	Wulff, C.; Ytterdal, T. "0.8V 1GHz Dynamic Comparator in Digital 90nm CMOS Technology" . Proceedings of NORCHIP 2005, © IEEE
	2005	Francisco Colodro, Marta Laguna, Carsten Wulff, Trond Ytterdal and Antonio Torralba. "New approach for Continuous Time Sigma Delta Modulators" . Proceedings of NORCHIP 2005
	2004	Wulff, C. Erstad, R. Ytterdal, T. "Programmable Analog Integrated Circuit" . Proceedings of NORCHIP 2004. © IEEE
	2002	Wulff, C. Ytterdal, T. "Programmable analog integrated circuit for use in remotely operated laboratories" . International Conference on Engineering Education (ICEE-2002). Visual supplements
	2002	Wulff, C. Ytterdal, T. Sæthre, T.A. Skjevlan, A. Fjeldy, T.A. et al "Next Generation Lab - A solution for remote characterization of analog integrated circuits" . International Caracas

Educational Work 2008 **Wulff, C;** Phd Thesis "[Efficient ADCs for nano-scale CMOS technology](#)" © Carsten Wulff
[Pipelined Schematics](#)
 2002 **Wulff, C;** Master Thesis "[Programmable Analog Integrated Circuit w/Table of Content](#)" © Carsten Wulff

Unpublished Works 2014 **Wulff, C;** "[The Universal Design Project](#)", 2014
 2008 **Wulff, C;** "[An ADC primer](#)", stuff about ADCs that did not make it into my PhD thesis. This is a rough cut, so there may be some trains of thought that might be difficult to follow.
 2006 **Wulff, C;** "[FE8113 High Speed Data Converters: A Summary](#)"
 2006 **Wulff, C;** "[Introduction to Mathematics of Noise Sources](#)"
 2006 **Wulff, C;** "[On Spectral Density](#)"
 2005 **Wulff, C;** "[A Brief Introduction to SystemDotNet](#)"

Patents 2018 [Reference voltages](#)
 2017 [Integrated circuits](#)
 2015 [Integrated power-on-reset circuit](#)
 2015 [Analogue-to-digital converter](#)

Co-Supervisor

Ph.D. students 2019-now Olivia Mirea
 2015-now Harald Garvik

Master Students 2004-2020 10+ students at NTNU

Hobbies

[Trailrunning and Bouldering](#)